Striver SDE Sheet - Linked List Part I

Quick DSA Revision Notes

Revision format: Approach -> core idea -> time and space complexity. Focus on the optimal pattern before interviews.

1. Reverse a Linked List

Approach	Core idea	Time	Space
Brute force	Store node values in a stack and overwrite them while popping. Links remain unchanged.	O(N)	O(N)
Optimal - Iterative	Use prev, curr and next. Save curr.next, reverse curr.next to prev, then move both pointers forward. prev becomes the new head.	O(N)	O(1)
Recursive	Reverse the remaining list, set head.next.next = head, then head.next = null.	O(N)	O(N) stack

Revision key: Key: Never change curr.next before saving the next node.

2. Find the Middle of a Linked List

Approach	Core idea	Time	Space
Brute force	Count nodes, then traverse N/2 steps from the head.	O(N)	O(1)
Optimal - Slow/Fast	Move slow by one and fast by two. When fast reaches the end, slow is at the middle. For even length, it returns the second middle.	O(N)	O(1)

Revision key: Pattern: Slow and fast pointers.

3. Merge Two Sorted Linked Lists

Approach	Core idea	Time	Space
Brute force	Put all values into an array, sort it and build a new list.	O((N+M) log(N+M))	O(N+M)
Better	Compare both lists and create a new sorted list.	O(N+M)	O(N+M)
Optimal	Use a dummy node and attach the smaller existing node each time. Append the remaining list at the end.	O(N+M)	O(1)

Revision key: Key: Dummy node removes special handling for the first node.

4. Remove Nth Node from End

Approach	Core idea	Time	Space
Brute force	Count total nodes, then delete the (N-n+1)th node from the beginning.	O(2N)	O(1)
Optimal - Two pointers	Use a dummy node. Move fast n steps ahead, then move fast and slow together until fast.next is null. Delete slow.next.	O(N)	O(1)

Revision key: Key: Dummy node safely handles deletion of the head.

5. Add Two Numbers

Approach	Core idea	Time	Space
Optimal	Traverse both lists together. sum = carry + available digits; digit = sum % 10; carry = sum / 10. Continue while a node or carry remains.	O(max(N,M))	O(max(N,M)) answer

Revision key: Key: Use a dummy node and do not forget the final carry.

6. Delete a Given Node

Approach	Core idea	Time	Space
Optimal	Copy the next node value into the current node, then bypass the next node: node.next = node.next.next.	O(1)	O(1)

Revision key: Limitation: The last node cannot be deleted using this method.